

Sums and products of zeroes of polynomials — Problem Set

Warm-up

1. Consider the quadratic equation $x^2 - 5x + 6 = 0$. Write down the value of the sum $(\alpha + \beta)$ and the product $(\alpha\beta)$ of the roots.
2. Given the cubic equation $x^3 + 4x^2 - 7x + 2 = 0$ with roots α, β, γ , find the value of $\alpha + \beta + \gamma$.
3. For the polynomial $P(x) = 2x^3 - 6x^2 + 4x - 1$, state the value of $\alpha\beta + \beta\gamma + \gamma\alpha$.
4. If α, β, γ are roots of $3x^3 + x^2 - 5 = 0$, find the product of the roots $\alpha\beta\gamma$.
5. A cubic equation has roots 2, 3, and -1 . Write the equation in the form $x^3 + bx^2 + cx + d = 0$.
6. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ for the equation $x^3 - 3x^2 + 2x - 5 = 0$.
7. If $\alpha + \beta + \gamma = 0$ and $\alpha\beta + \beta\gamma + \gamma\alpha = -7$, find the value of $\alpha^2 + \beta^2 + \gamma^2$.
8. Identify the coefficients a, b, c, d for the equation $4x^3 - x + 2 = 0$ and find the sum of the roots.
9. For $P(x) = x^3 - 2x^2 + kx - 8$, find k if the product of two roots is 4.
10. If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$, express $(\alpha + 1)(\beta + 1)(\gamma + 1)$ in terms of p, q, r .

Skill-building

1. The roots of $2x^3 - 8x^2 + 5x - 3 = 0$ are α, β, γ . Find the value of $\alpha^2 + \beta^2 + \gamma^2$.
2. Given $x^3 + px^2 + qx + r = 0$ has roots α, β, γ , show that $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = -pr$.
3. The equation $x^3 - 7x^2 + 14x - 8 = 0$ has roots in arithmetic progression. Find the roots.
4. If α, β, γ are roots of $x^3 + 2x^2 + 3x + 4 = 0$, find the value of $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$.
5. The polynomial $P(x) = x^3 - 6x^2 + ax - 4$ has real positive roots. If one root is the product of the other two, find a .
6. Find the cubic equation whose roots are the squares of the roots of $x^3 - x - 1 = 0$.
7. If α, β, γ are roots of $x^3 + px + q = 0$, find the value of $\alpha^3 + \beta^3 + \gamma^3$ in terms of q .
8. For the equation $3x^3 - 9x^2 + 6x - 2 = 0$, find the value of $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$.
9. The roots of $x^3 - 12x^2 + 39x - 28 = 0$ are in geometric progression. Find the roots.
10. Given α, β, γ are roots of $x^3 + bx^2 + cx + d = 0$, find the value of $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\gamma\alpha} + \frac{\gamma}{\alpha\beta}$.

Easier Exam Questions

1. (Sydney Grammar 2019 13a)

The polynomial $P(x) = x^3 + 8x^2 + cx - 48$ has a zero that is the sum of the other two. That is, the zeros of $P(x)$ are α, β , and $\alpha + \beta$.

- (i) Write down the value of $\alpha + \beta$. 1
- (ii) Find the value of c . 1
- (iii) Fully factorise $P(x)$. 2

2. (Glenwood 2022 Q9b)

If α, β, γ are the roots of the equation $2x^3 - 6x^2 + 10x - 5 = 0$, find the values of:

- i) $\alpha + \beta + \gamma$ 1
- ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
- iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

3. (Baulkham Hills 2024 Q11d)

If α, β and γ are roots of $2x^3 - 5x^2 + 3x - 5 = 0$, find the value of:

- (i) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
- (ii) $\frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\beta}$. 2

4. (Glenwood 2020 Q8a)

a) If α, β and γ are the roots of the cubic equation $5x^3 + 2x^2 - x - 3 = 0$, find the value of

- i) $\alpha + \beta + \gamma$ 1
- ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1
- iii) $\alpha^2 + \beta^2 + \gamma^2$ 2
- iv) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$ 1

5. (Penrith 2024 Q13b)

The roots of $x^3 + 5x^2 + 7x - 1 = 0$ are α, β and γ .

- (i) Find $\alpha(\beta + 1) + \beta(\gamma + 1) + \gamma(\alpha + 1)$. 2
- (ii) Find $\alpha^2 + \beta^2 + \gamma^2$. 1

6. (Glenwood 2019 Q11)

The equation $x^3 - 6x^2 + 4x - 3 = 0$ has roots α, β and γ .

Find the value of $\alpha^2 + \beta^2 + \gamma^2$ 2

7. (Presbyterian 2023 Q8)

Let $P(x) = ax^3 + bx^2 + cx + a$ where a and b are constants and $a \neq 0$.

The three distinct zeroes are $-1, \alpha$ and β .

Which of the following is a true statement?

- (A) $\beta = -\frac{1}{\alpha}$
- (B) $\beta = \frac{1}{\alpha}$
- (C) $\beta = \alpha - 1$

(D) $\beta = 1 - \alpha$

8. (Baulkham Hills 2025 Q11e)
 If α, β and γ are roots of $2x^3 - 4x^2 + 1 = 0$, find the value of:
- i) $\alpha^2 + \beta^2 + \gamma^2$ 2
- ii) $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$ 1
9. (North Sydney Boys 2023 Q3)
 α, β and γ are solutions of $2x^3 - 4x^2 - 3x - 1$,
- i) find the value of $\alpha\beta + \beta\gamma + \alpha\gamma$ 1
- ii) find the value of $(\alpha - 1)(\beta - 1)(\gamma - 1)$ 2
10. (Sydney Girls 2021 Q2b)
 Given α, β and γ are the roots of $2x^3 + mx^2 + x + 3 = 0$:
- (i) Find the value of m given the sum of the roots is 5. 1
- (ii) Evaluate $(\alpha - 1)(\beta - 1)(\gamma - 1)$. 2

Harder Exam Questions

1. (Fort St 2022 Q10b)
 The three roots of $x^3 + px + qx + r = 0$ are integers with a common difference of β . That is, if the roots are x_1, x_2 and x_3 then $x_2 - x_1 = x_3 - x_2 = \beta$. Show that $9pq = 27r + 2p^3$. 3
2. (Fort St 2019 Q1c)
 The polynomial $P(x) = x^3 + px^2 + qx + r$ has roots $0, 2, -2$.
- i) Find the values of p, q and r . 2
- ii) Sketch the graph of $y = P(x)$, showing any intercepts with the coordinate axes. 2
- iii) Hence, or otherwise, solve $\frac{x^2-4}{x} \geq 0$ 2
3. (Hurlstone 2024 Q15c)
 If α, β, γ are the roots of the equation $2x^3 - 5x^2 - 3x + 1 = 0$,
 find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$. 2
4. (Baulkham Hills 2019 Q13a)
 Let the cubic polynomial $P(x) = x^3 - 3x^2 + kx + 12$ have roots α, β and γ .
- (i) Find the value of $\alpha + \beta + \gamma$ 1
- (ii) Find the value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$ 2
- (iii) Given that two of its roots have a sum of zero, find the value of k 2
5. (Sydney Tech 2019 Q16b)
 The roots of the equation $x^3 - 3x^2 + 4x + 2 = 0$ are α, β, γ . Find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$
2
6. (Moriah 2021 Q7a)
 The roots of the cubic equation $x^3 - 3x^2 + 4x + 2 = 0$ are α, β, γ . 2
 Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

7. (Baulkham Hills 2022 Q14a)
 The roots of the polynomial $x^3 - x^2 - 5x + 2 = 0$ are α, β and γ .
- (i) Write down the values of $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$. **2**
- (ii) Show that $\beta + \gamma = 1 - \alpha$. **1**
- (iii) Hence evaluate $\frac{\alpha+\beta}{\gamma} + \frac{\beta+\gamma}{\alpha} + \frac{\gamma+\alpha}{\beta}$. **2**
8. (Glenwood 2024 Q9e)
 If α, β, γ are roots of $3x^3 + 8x^2 - 1 = 0$, find the value of $\left(\beta + \frac{1}{\gamma}\right) \left(\gamma + \frac{1}{\alpha}\right) \left(\alpha + \frac{1}{\beta}\right)$. **3**
 (Do not find the values of α, β and γ)
9. (Blacktown Boys 2020 Q13c)
 $P(x) = x^3 - 6x^2 + ax - 4$ where $a > 0$. Given that all the roots of $P(x)$ are real and positive, and that one of the roots is the product of the other two roots, find the value of a . **3**
10. (Hurlstone 2019 Q6e)
 If $\tan A$ and $\tan B$ are the roots of the equation $3x^2 - 5x - 1 = 0$, find the value of $\tan(A + B)$ **2**
11. (Gosford 2020 Q10b)
 The equation $2x^3 + 6x^2 + 4x + 1 = 0$ has roots α, β and γ .
- (i) Find the value of $(1 - \alpha) + (1 - \beta) + (1 - \gamma)$. **1**
- (ii) Find the value of $(1 - \alpha)(1 - \beta)(1 - \gamma)$. **2**
12. (Girraween 2024 Q7e)
 The zeros of $x^3 - 5x + 3 = 0$ are α, β and γ , find a cubic equation with integer coefficients whose zeros are $2\alpha, 2\beta$ and 2γ . **4**